Neural Networks Overview

Neural networks are machine-learning models inspired loosely by the way networks of neurons process information in biological brains. They contain layers of connected units that receive numerical inputs, transform them, and pass results to later layers. During training, the network compares its predictions with known answers and adjusts internal values called weights to reduce error. This process allows a neural network to learn patterns that may be difficult to describe using fixed rules. Neural networks are widely used for image classification, speech recognition, text generation, translation, and game-playing systems. Deep learning refers to neural networks with many layers, which can learn increasingly complex representations from large amounts of data. Their performance can be impressive, but training may require substantial data, computing power, and careful evaluation to avoid overfitting or unreliable outputs.